4,212,1 Macroeconomics III
Exercises and Independent Studies

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Outline

- The Solow Model for a Small Open Economy (Exercise 2.1)
- The General Solow Model (Exercise 2.2)

The Solow Model for a Small Open Economy

- Closed Economy:

$$S_t - I_t = 0 \Longleftrightarrow \frac{S_t}{Y_t} = \frac{I_t}{Y_t}$$

- Open Economy:

$$S_t - I_t = F_{t+1} - F_t = CA_t$$

Exercise 2.1

The Solow Model for a Small Open Economy:

$$
\begin{aligned}
Y_t &= B(K_t)^{\alpha}(L_t)^{1-\alpha}, \\
V_t &= K_t + F_t, \\
V_{t+1} - V_t &= S_t, \\
Y_t^n &= Y_t + \bar{r}F_t, \\
w_t &= (1-\alpha)B(K_t)^{\alpha}L_t^{-\alpha}, \\
\bar{r} &= \alpha B(K_t)^{\alpha-1}(L_t)^{1-\alpha}, \\
C_t &= (1-s)Y_t^n, \\
S_t &= sY_t^n, \\
L_{t+1} &= (1+n)L_t.
\end{aligned}
$$

4 Parameters: $\bar{r}, \alpha, s, n$. 9 Endogenous variables: Y, Y^n, K, L, w, S, F, C, and V of which V_t and L_t are state (predetermined) variables.

Transformations in per Worker Terms

The set of equations with transformed variables in per worker terms is:

$$
\begin{aligned}
&y_t = B(k_t)^\alpha, \\
&v_t = k_t + f_t, \\
&y_t^n = y_t + \bar{r} f_t = w_t + \bar{r} v_t, \\
&v_{t+1} = v_t + s_t = \frac{1 + s\bar{r}}{1+n} v_t + \frac{s w_t}{1+n}, \\
&w_t = (1-\alpha) B(k_t)^\alpha, \\
&\bar{r} = \alpha B(k_t)^{\alpha-1}, \\
&c_t = (1-s) y_t^n, \\
&s_t = s y_t^n, \\
&\overline{\rho} = \bar{r},\ \delta = 0.
\end{aligned}
$$

Note that

$$
s_t - i_t = f_{t+1}(1+n) - f_t.
$$

Steady State

Set $v_{t+1} = v_t = v^*$. This implies that:

$$v^* = \frac{s/n}{1 - s\bar{r}/n} w^*,$$

$$k^* = (\frac{\bar{r}}{\alpha B})^{\frac{1}{\alpha - 1}},$$

$$w^* = (1 - \alpha) B (k^*)^{\alpha},$$

$$f^* = \frac{1}{1 - \alpha} \frac{s}{n} \frac{1}{\bar{r}} \frac{\bar{r} - \alpha n/s}{1 - s\bar{r}/n} w^*,$$

$$y^* = B(k^*)^{\alpha},$$

$$y^{n^*} = \frac{1}{1 - s\bar{r}/n} w^*,$$

$$c^* = (1 - s) y^{n^*},$$

$$s^* = s y^{n^*},$$

$$i^* = s^* - n f^*.$$

Autarky versus Free Capital Mobility

- Autarky (closed economy): $r_c^* = \frac{\alpha n}{s}$. World economy: $\bar{r} = \frac{\alpha \bar{n}}{\bar{s}}$.
- Case 1: $r_c^* < \bar{r}$. The economy has a relatively strong propensity to save ($\frac{s}{n} > \frac{\bar{s}}{\bar{n}}$), which implies capital accumulation is relatively strong and the real interest rate is relatively low under autarky. Opening the capital account: the owners of capital take advantage of a higher international interest rate $\rightarrow$ capital flows out of the domestic economy $\rightarrow$ capital intensity decreases $\rightarrow$ higher MP of capital, lower MP of labor $\rightarrow$ domestic interest rate increases to $\bar{r}$, domestic real wages fall.
- Case 2: $r_c^* > \bar{r}$. The economy has a relatively low propensity to save ($\frac{s}{n} > \frac{\bar{s}}{\bar{n}}$) $\rightarrow$ capital accumulation is relatively weak and the real interest rate is relatively high under autarky. An opening creates an arbitrage possibility: borrow on the international capital market and invest in domestic capital, capital flows in $\rightarrow$ capital intensity increases $\rightarrow$ MP of capital decreases, MP of labor increases $\rightarrow$ domestic interest rate falls to $\bar{r}$ and domestic real wage increases.

Autarky versus Free Capital Mobility

		Autarky	Open Economy
Capital	k^*	11.18	37.04
GDP	y^*	2.24	3.33
Consumption	c^*	1.90	2.22
Investment	i^*	0.34	1.11
Interest rate	r^*	0.07	0.03
Wage rate	w^*	1.49	2.22
Net foreign assets	f^*	0.00	-23.97
National income	y^{n^*}	2.24	2.61
National wealth	v^*	11.18	13.07
Interest payments	$\bar{r}f^*$	0.00	-0.72

- The stability condition, $s\bar{r} < n \Leftrightarrow 0.15 \times 0.3 < 0.3$ is satisfied.
- The table indicates substantial increases in capital and domestic production, resulting from capital inflows.
- As domestic production is partly used to pay off interest, the increase in national income and consumption is more modest.

Exercise 2.2

Assume: $F(K_t, L_t) = (K_t)^\alpha (A_t L_t)^{1-\alpha}$. Then:

$$
\begin{aligned}
Y_t &= (K_t)^\alpha (A_t L_t)^{1-\alpha}, \\
A_{t+1} &= (1+g) A_t, \\
K_{t+1} &= S_t + (1-\delta) K_t, \\
w_t &= (1-\alpha)(K_t)^\alpha L_t^{-\alpha} A_t^{1-\alpha}, \\
r_t &= \alpha (K_t)^{\alpha-1} (A_t L_t)^{1-\alpha}, \\
C_t + S_t &= Y_t, \\
S_t &= s Y_t, \\
L_{t+1} &= (1+n) L_t, \\
\rho_t &= r_t - \delta.
\end{aligned}
$$

5 Parameters: $g, \alpha, s, n, and \delta$. 9 Endogenous variables: Y, K, L, r, w, S, C, A, and ρ of which K_t, A_t and L_t are state (predetermined) variables.

Transformations in per 'Efficiency Unit of Labor' Terms

Define: $\tilde{k}_t = \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}$ and $\tilde{y}_t = \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}$. Then:

$$\frac{K_{t+1}}{A_{t+1} L_{t+1}} = s \frac{Y_t}{L_t(1+n) A_t (1+g)} + (1-\delta) \frac{K_t}{L_t(1+n) A_t (1+g)},$$

which is equivalent to

$$\tilde{k}_{t+1} = s \frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{k}_t}{(1+n)(1+g)}.$$

Transformations in per 'Efficiency Unit of Labor' Terms

The set of equations with transformed variables in per efficiency unit of labor terms is:

$$
\begin{aligned}
&\tilde{y}_t = (\tilde{k}_t)^{\alpha}, \\
&\tilde{k}_{t+1} = s\frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta)\frac{\tilde{k}_t}{(1+n)(1+g)}, \\
&\tilde{w}_t = (1-\alpha)(\tilde{k}_t)^{\alpha}, \\
&r_t = \alpha(\tilde{k}_t)^{\alpha-1}, \\
&\tilde{c}_t + \tilde{s}_t = \tilde{y}_t, \\
&\tilde{s}_t = s\tilde{y}_t, \\
&\rho_t = r_t - \delta.
\end{aligned}
$$

The Transition Equation

Replacing the production function: $\tilde{y}_t = (\tilde{k}_t)^\alpha$ in the capital accumulation equation, we obtain the transition equation:

$$\tilde{k}_{t+1} = s\frac{(\tilde{k}_t)^\alpha}{(1+n)(1+g)} + (1-\delta)\frac{\tilde{k}_t}{(1+n)(1+g)}.$$

Set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$:

$$\tilde{k}^* = \left(\frac{s}{n+g+\delta+ng}\right)^{\frac{1}{(1-\alpha)}}$$

Steady State

$$\tilde{k}^* = \left(\frac{s}{n + g + \delta + ng}\right)^{\frac{1}{(1-\alpha)}}$$

$$\tilde{y}^* = \left(\frac{s}{n + g + \delta + ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$\tilde{w}^* = (1 - \alpha)\left(\frac{s}{n + g + \delta + ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$r^* = \alpha\left(\frac{s}{n + g + \delta + ng}\right)^{-1}$$

$$\tilde{c}^* = (1 - s)\left(\frac{s}{n + g + \delta + ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$\tilde{s}^* = s\left(\frac{s}{n + g + \delta + ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$\rho^* = \alpha\left(\frac{s}{n + g + \delta + ng}\right)^{-1} - \delta$$

Steady State

Calibration: $\alpha = \frac{1}{3}$, $n = 0.01$, $\delta = 0.10$, $g = 0.02$, $s = 0.45$ in Economy 1 and $s = 0.35$ in Economy 2.

Table: Comparison of Economies

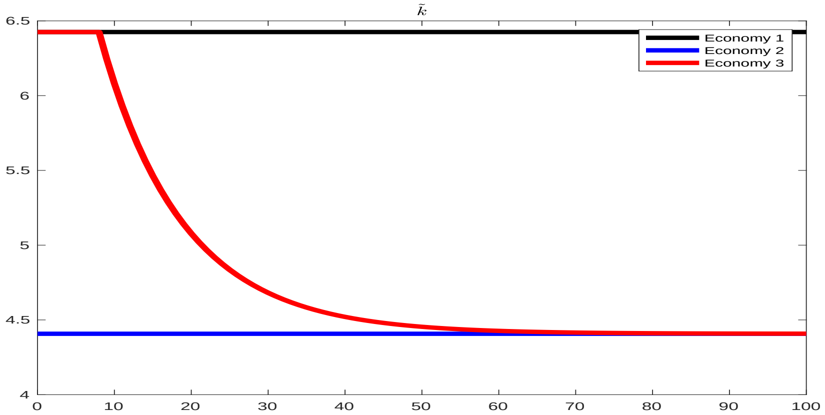
	Economy 1 (s=0.45)	Economy 2 (s=0.35)
$\tilde{k}^*$	6.43	4.41
$\tilde{y}^*$	1.86	1.64
$\tilde{c}^*$	1.02	1.07

- $\tilde{k}^*$, $\tilde{y}^*$ decrease if s is reduced.
- $\tilde{c}^*$ increases if s is reduced, given that $s > 1/3$. Why?

Variables in per 'Worker' Terms

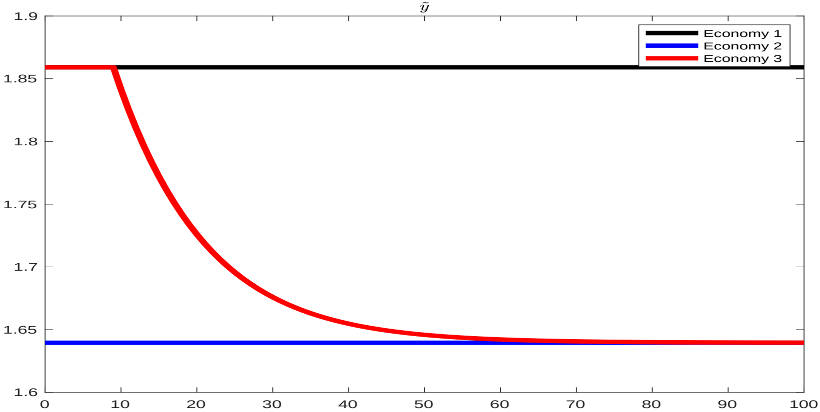
$$A_t = (1+g)^t A_0$$
$$k_t = \tilde{k}_t A_t$$
$$\ln(y_t) = \ln(\tilde{y}_t) + \ln(A_t)$$
$$\ln(c_t) = \ln(\tilde{c}_t) + \ln(A_t)$$
$$g_t^y = \ln(y_t) - \ln(y_{t-1})$$
$$\frac{k_t}{y_t} = \frac{\tilde{k}_t}{\tilde{y}_t}$$

Decrease in the Savings Rate



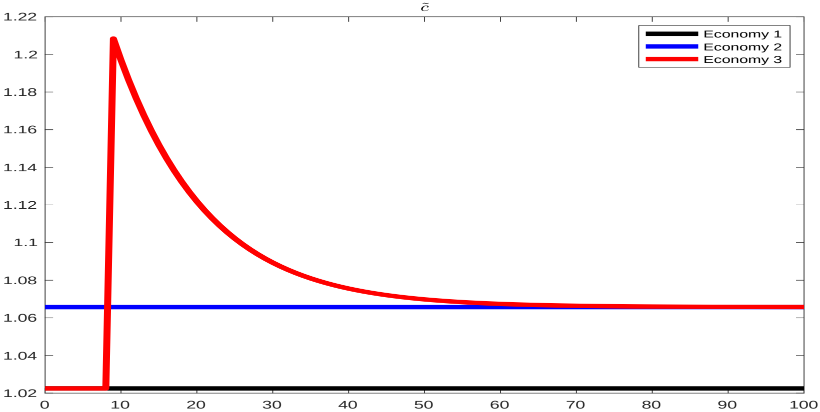
Period 10: A smaller fraction of an unchanged level of income is being saved. The capital stock decreases starting in period 11 until it reaches the new steady state.

Decrease in the Savings Rate



Period 10: The level of income is unchanged. It starts decreasing in period 11 because capital decreases and continues decreasing until it reaches the new steady state.

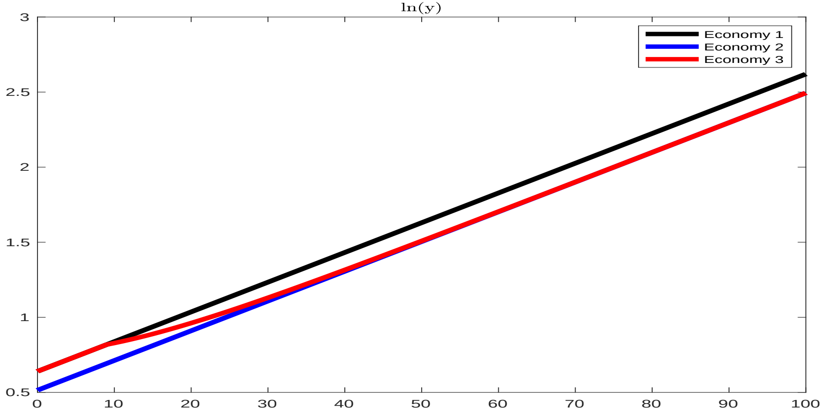
Decrease in the Savings Rate



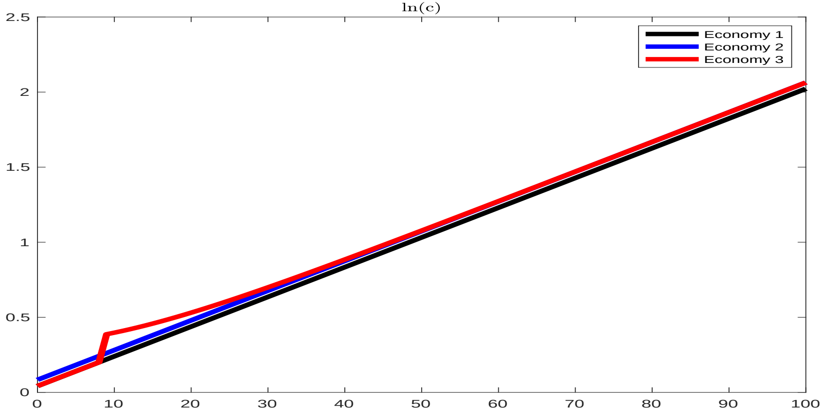
Decrease in the Savings Rate

- A decrease in s always results in an immediate increase in c_t in the short run because a smaller fraction of an unchanged level of income is being saved.
- After the initial short-run increase, c_t starts to decrease as the capital stock decreases, and hence income decreases.
- Whether the economy ends up with more or less consumption in the long run depends on where s was initially and where s ends up relative to the Golden Rule.

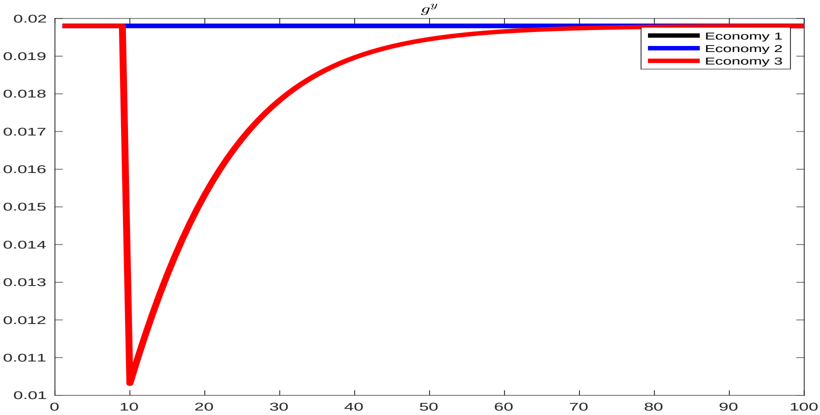
Decrease in the Savings Rate



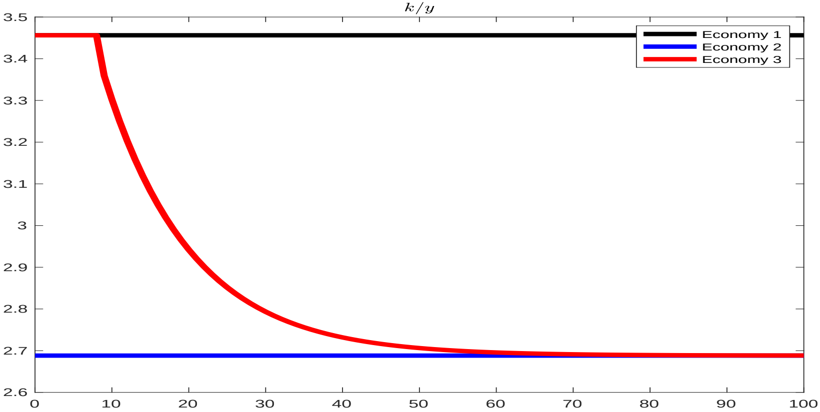
Decrease in the Savings Rate



Decrease in the Savings Rate



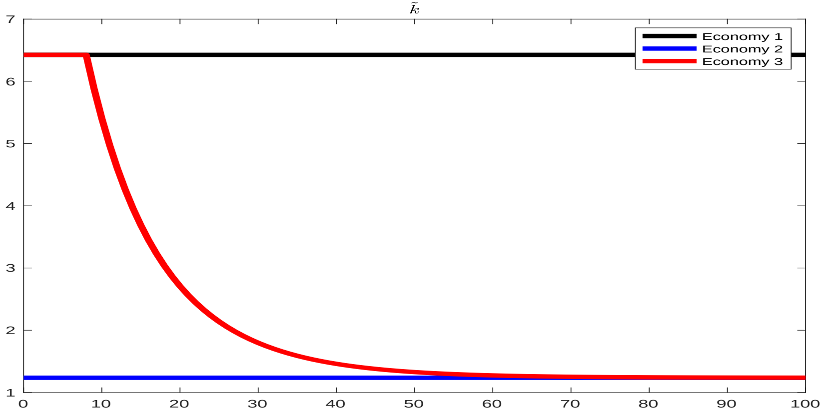
Decrease in the Savings Rate



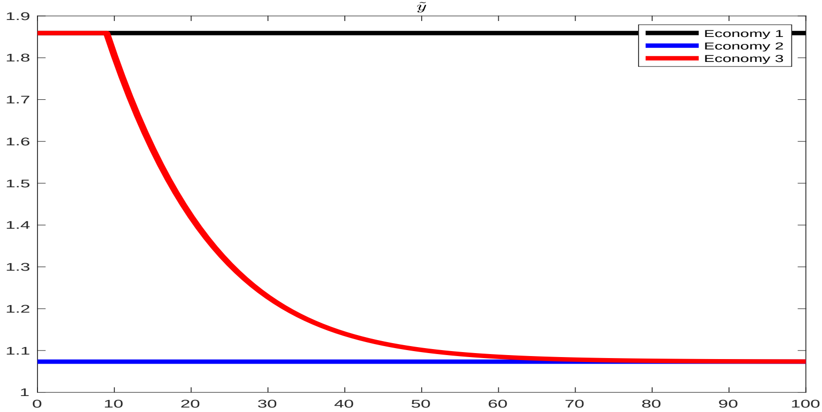
Decrease in the Savings Rate

What happens if the savings rate drops from 0.45 to 0.15?

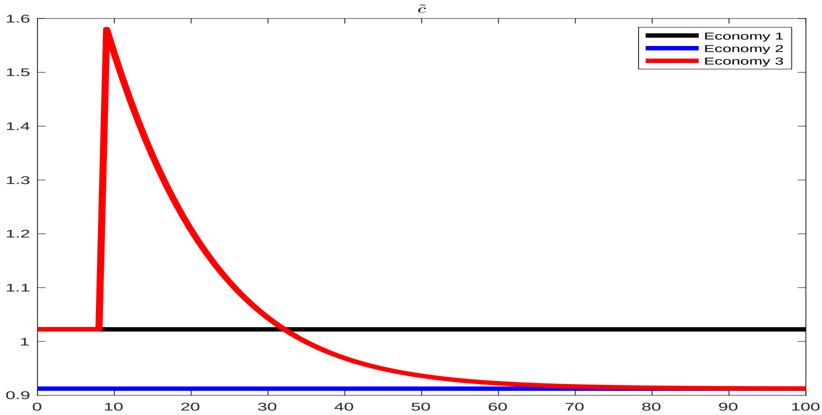
Decrease in the Savings Rate



Decrease in the Savings Rate



Decrease in the Savings Rate



Decrease in the Savings Rate

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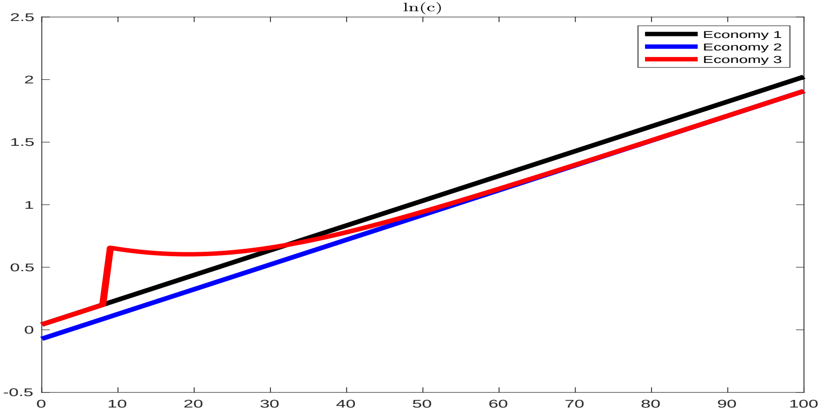
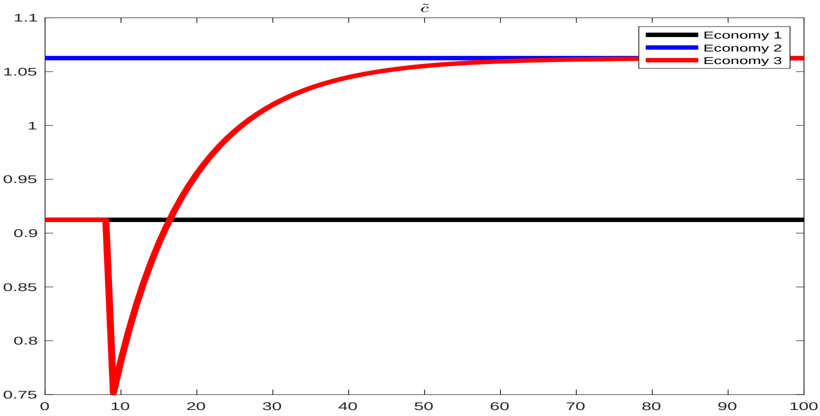
Decrease in the Savings Rate



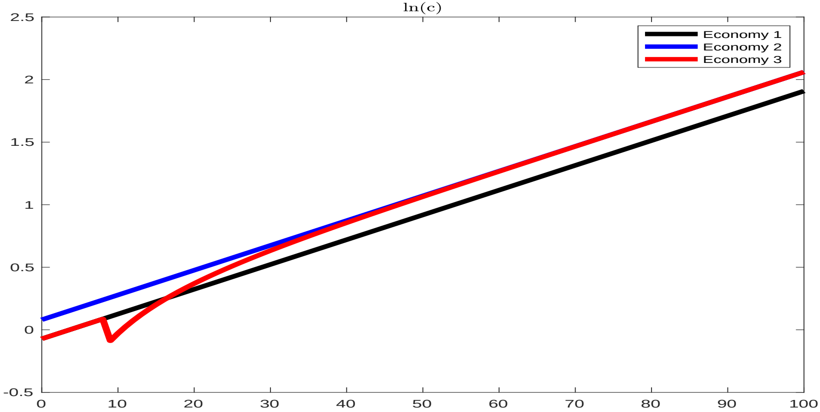
Increase in the Savings Rate

What happens if the savings rate increases from 0.15 to 0.3?

Increase in the Savings Rate



Increase in the Savings Rate



Increase in the Savings Rate

Would you like that?

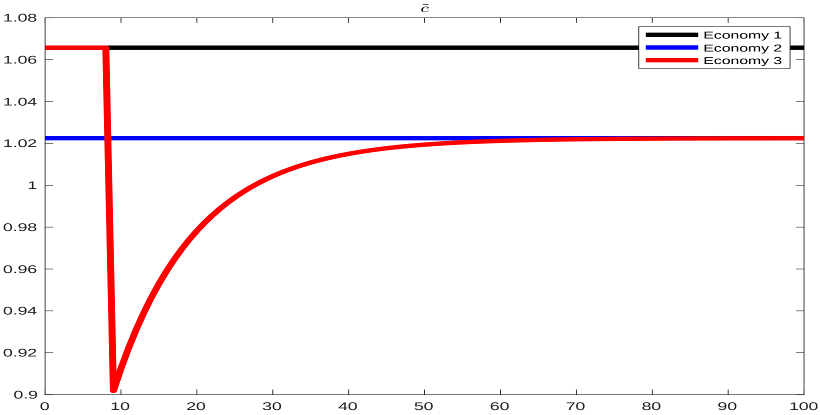
Increase in the Savings Rate

- Not necessarily.
- The higher long-run consumption is only achieved through lower consumption in the short run.
- Are you willing to endure the short-run pain for the long-run gain?
- It depends on how impatient you are.
- Moreover, if the dynamics are sufficiently slow, those who sacrificed by consuming less in the present may be dead by the time the average level of consumption increases.
- Do you care more about future generations than about you?
- Without saying something more specific about how impatient people are or how they value the well-being of future generations, it is not possible to conclude whether or not the saving rate should increase.

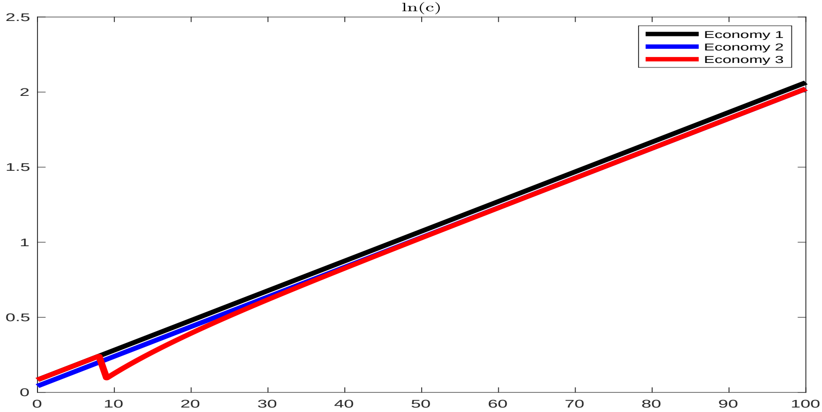
Increase in the Savings Rate

What happens if the savings rate increases from 0.35 to 0.45?

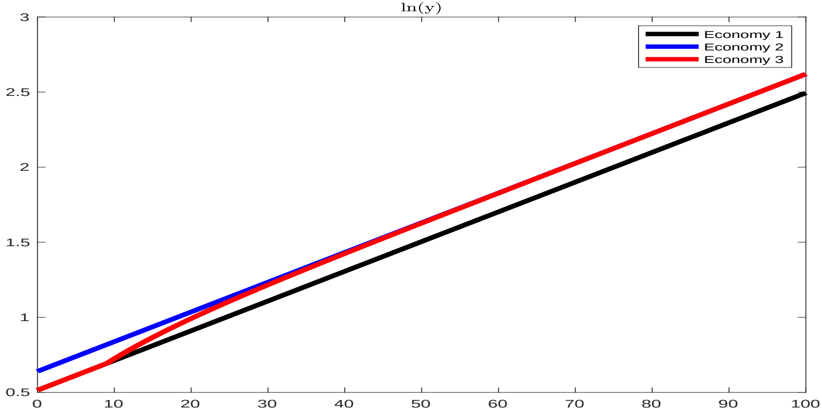
Increase in the Savings Rate



Increase in the Savings Rate



Increase in the Savings Rate



Conclusion about the Savings Rate

- If initially $s < s^{gr}$, then a **small** increase in s results in a long-run increase in consumption in the new steady state.
- If initially $s > s^{gr}$, then an increase in s results in a long-run decrease in consumption in the new steady state.
- An economy with $s > s^{gr}$ is "dynamically inefficient" because the economy is saving too much and consumption is being left on the table both in the present and in the future.
- In this case, households would be unambiguously better off by reducing s to values close to s^{gr}. Regardless of how impatient people are, a reduction in s gives people more consumption at every date, and this makes them better off. However, if the reduction is big and s goes well below s^{gr}, consumers will be left with less consumption in the long run.